Radicals & Exponents

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Multiplying polynomials**

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(5)$.

Final Answer: 31. Substitute $x = 5$: $f(5) = 5(5) + 6 = 31$.

2 **[Foundational]** • **Target: Multiplying polynomials**

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

Final Answer: $x = 4$ and $x = 7$. Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

3 **[Foundational]** • **Target: Special products**

For the model $f(x) = 6x + 7$ in special products, evaluate $f(2)$.

Final Answer: 19. Substitute $x = 2$: $f(2) = 6(2) + 7 = 19$.

4 **[Foundational]** • **Target: Special products**

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

Final Answer: $x = 1$ and $x = 5$. Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

5 **[Foundational]** • **Target: Special products**

A data set recorded for special products consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

6 **[Foundational]** • **Target: Special products**

For the model $f(x) = 6x + 7$ in special products, evaluate $f(3)$.

Final Answer: 25. Substitute $x = 3$: $f(3) = 6(3) + 7 = 25$.

7 **[Foundational]** • **Target: Special products**

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

Final Answer: $x = 2$ and $x = 6$. Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Special products

For the model $f(x) = 6x + 7$ in special products, evaluate $f(4)$.

Final Answer: 31. Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

9 [Intermediate] • Target: Special products

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

Final Answer: $x = 3$ and $x = 7$. Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

10 [Intermediate] • Target: Special products

For the model $f(x) = 6x + 7$ in special products, evaluate $f(5)$.

Final Answer: 37. Substitute $x = 5$: $f(5) = 6(5) + 7 = 37$.

11 [Intermediate] • Target: Special products

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

12 [Intermediate] • Target: Factoring out the greatest common factor (GCF)

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(2)$.

Final Answer: 12. Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

13 [Intermediate] • Target: Factoring out the greatest common factor (GCF)

A data set recorded for factoring out the greatest common factor (gcf) consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 19.5. The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

14 [Intermediate] • Target: Factoring out the greatest common factor (GCF)

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(3)$.

Final Answer: 14. Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Factoring out the greatest common factor (GCF) • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(4)$.

Final Answer: 16. Substitute $x = 4$: $f(4) = 2(4) + 8 = 16$.

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16 [Challenging] • Target: Factoring out the greatest common factor (GCF) • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(5)$.

Final Answer: 18. Substitute $x = 5$: $f(5) = 2(5) + 8 = 18$.

17 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

18 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 6.01 Systems & Inequalities

A data set recorded for factoring trinomials consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

19 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(3)$.

Final Answer: 18. Substitute $x = 3$: $f(3) = 3(3) + 9 = 18$.

20 [Challenging] • Target: Factoring trinomials • Cross-Lesson: 8.01 Factoring Polynomials

For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(4)$.

Final Answer: 21. Substitute $x = 4$: $f(4) = 3(4) + 9 = 21$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

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22 [Foundational] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

23 [Foundational] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(5)$.

a. 19 ✓

b. 22

c. 15

d. 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

24 [Foundational] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

25 [Foundational] • Target: Multiplying a polynomial by a monomial

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

26 [Foundational] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(5)$.

a. 25 ✓

b. 29

c. 20

d. 29

Substitute $x = 5$: $f(5) = 4(5) + 5 = 25$.

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27 [Foundational] • Target: Multiplying polynomials

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(2)$.

a. 16 ✓

b. 21

c. 10

d. 20

Substitute $x = 2$: $f(2) = 5(2) + 6 = 16$.

28 [Foundational] • Target: Multiplying polynomials

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

30 [Intermediate] • Target: Adding and subtracting polynomials

A data set recorded for adding and subtracting polynomials consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

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31 [Intermediate] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

32 [Intermediate] • Target: Adding and subtracting polynomials

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(4)$.

a. 16 ✓

b. 19

c. 12

d. 20

Substitute $x = 4$: $f(4) = 3(4) + 4 = 16$.

33 [Intermediate] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

34 [Intermediate] • Target: Adding and subtracting polynomials

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ **and** $x = 9$ ✓

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

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35 [Intermediate] • Target: Multiplying a polynomial by a monomial

A data set recorded for multiplying a polynomial by a monomial consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

36 [Intermediate] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(3)$.

a. 17 ✓

b. 21

c. 12

d. 21

Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

37 [Intermediate] • Target: Multiplying a polynomial by a monomial

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(4)$.

a. 21 ✓

b. 25

c. 16

d. 25

Substitute $x = 4$: $f(4) = 4(4) + 5 = 21$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiplying a polynomial by a monomial • Cross-Lesson: 6.01 Systems & Inequalities

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

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39 [Challenging] • Target: Multiplying a polynomial by a monomial • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

40 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

41 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 1.01 Linear Equations & Lines

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(3)$.

a. 21 ✓

b. 26

c. 15

d. 25

Substitute $x = 3$: $f(3) = 5(3) + 6 = 21$.

42 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

43 [Challenging] • Target: Multiplying polynomials • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

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44 [Challenging] • Target: Special products • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for special products consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

45 [Challenging] • Target: Factoring out the greatest common factor (GCF) • Cross-Lesson: 3.01 Exponents & Polynomials

A data set recorded for factoring out the greatest common factor (gcf) consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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